Introduction to Loops

Understanding Iteration using Flowcharts

Programming Fundamentals Team

SETU

2026-06-18

Outline

Outline

1.1 What if we needed to repeat this?

[Without loops, repeated tasks quickly become messy:

```
print("Welcome")
print("Welcome")
print("Welcome")
print("Welcome")
```

A loop lets us write the repeated instruction once.],

1.1 What if we needed to repeat this?

Key idea

Computers are very good at repetition.

1.2 Everyday repetition

- Checking each student's mark
- Taking orders until the queue is empty
- Asking for a password until it is correct
- Displaying a menu until the user chooses Exit
- Reading temperatures from a sensor

Loops are used when a set of instructions must be repeated.

1.3 What is iteration?

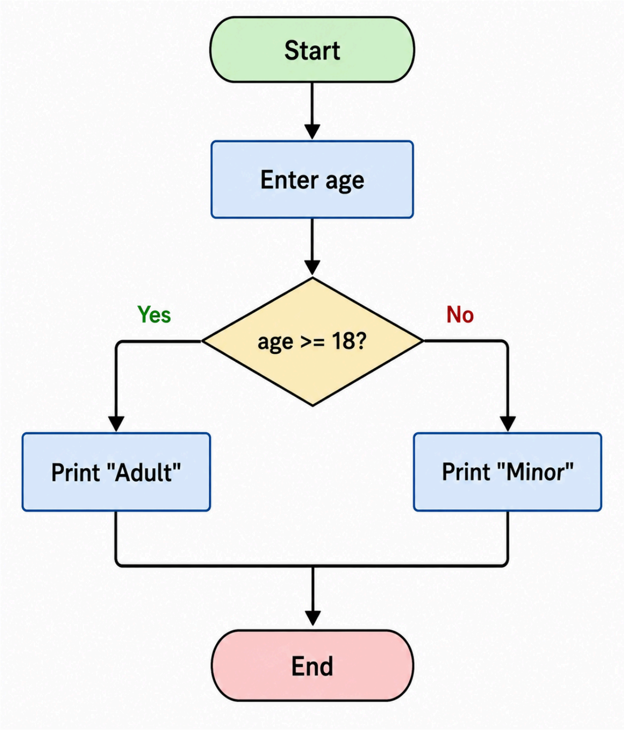
Iteration means repeating a set of instructions until a condition is met.

Examples:

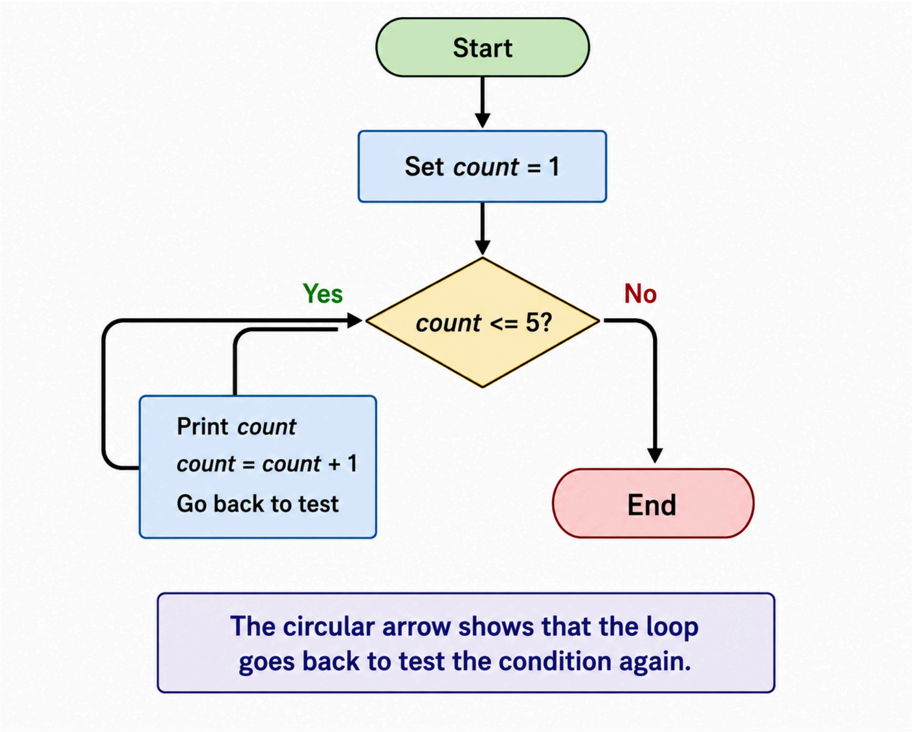
- Continue asking until the input is valid.
- Continue counting until the target is reached.
- Continue processing until there is no more data.

Outline

2.1 Selection asks once



2.2 Iteration asks repeatedly



2.3 The key difference

Selection

- Condition checked once
- One path is chosen
- Program continues

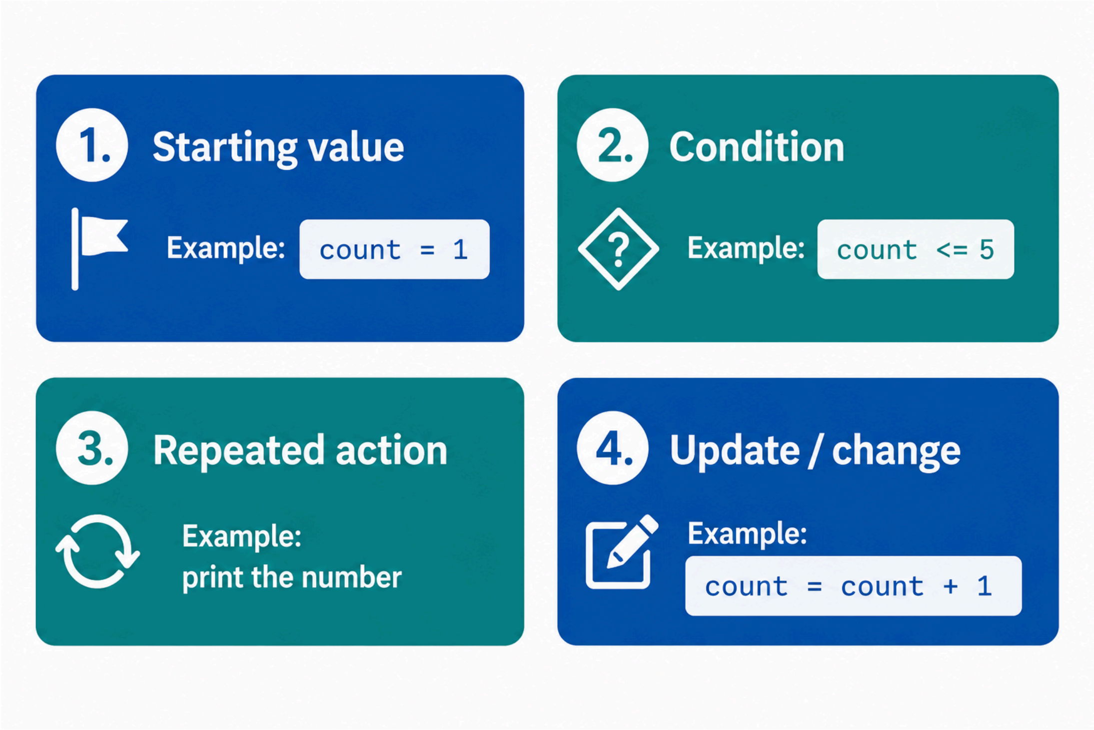
2.3 The key difference

Iteration

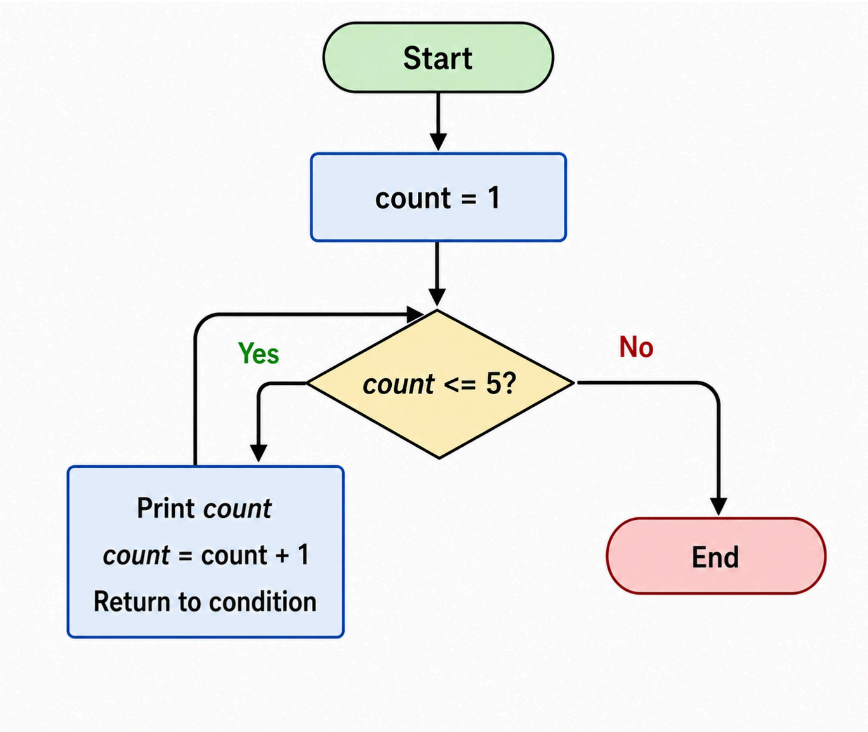
- Condition checked repeatedly
- Instructions may run many times
- Stops when condition becomes false

Outline

3.1 Every loop needs four things



3.2 Counting to five: flowchart



3.3 Trace the loop

count	Condition	Output
1	1 <= 5 is True	1
2	2 <= 5 is True	2
3	3 <= 5 is True	3
4	4 <= 5 is True	4
5	5 <= 5 is True	5
6	6 <= 5 is False	Stop

3.4 Why does the loop stop?

The loop stops because the value changes.

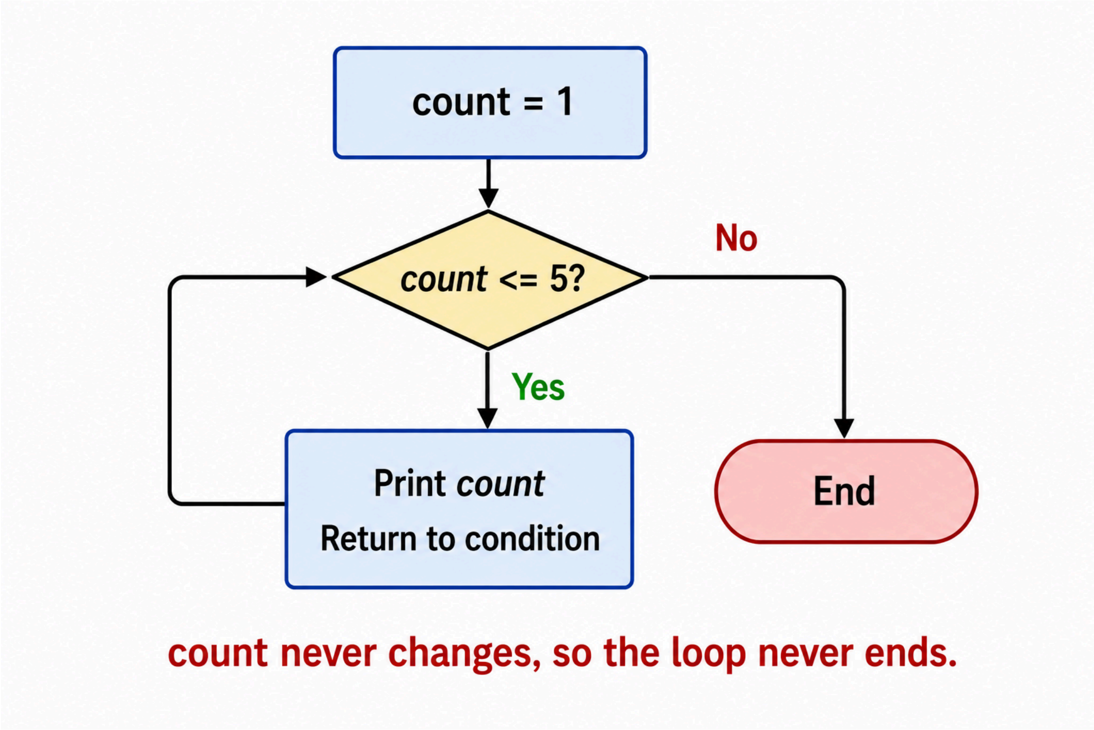
`count` eventually becomes 6.

The condition `count <= 5` becomes false.

Important: a loop normally needs something inside it that moves it closer to stopping.

Outline

4.1 What if we forget the update?



4.2 Infinite loop in code

```
count = 1

while count <= 5:
    print(count)
```

Problem: `count += 1` is missing.

4.3 Corrected code

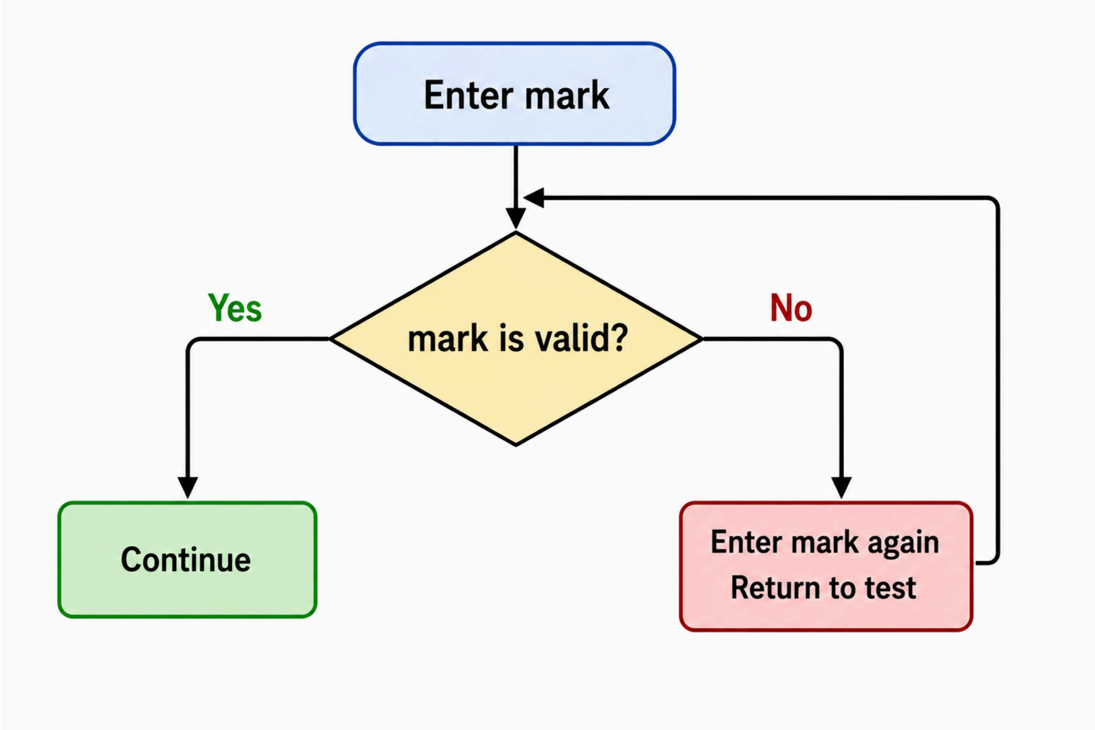
```python
count = 1

while count <= 5:
    print(count)
    count += 1
```

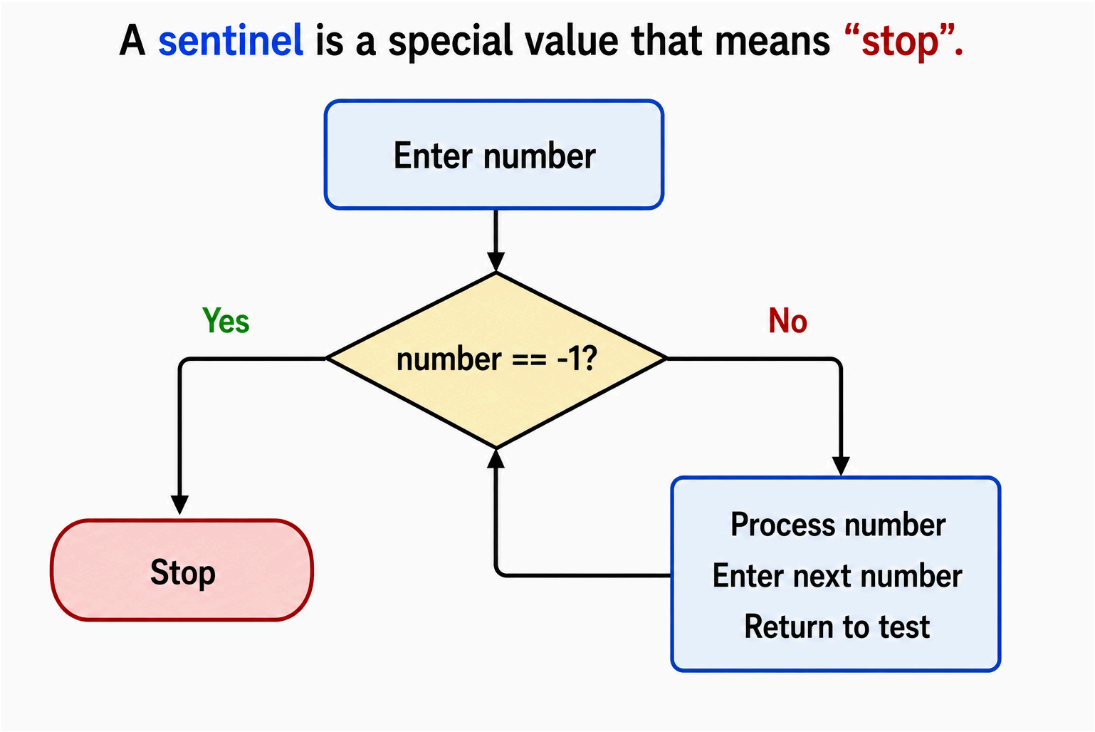
Outline

5.1 Validation loop

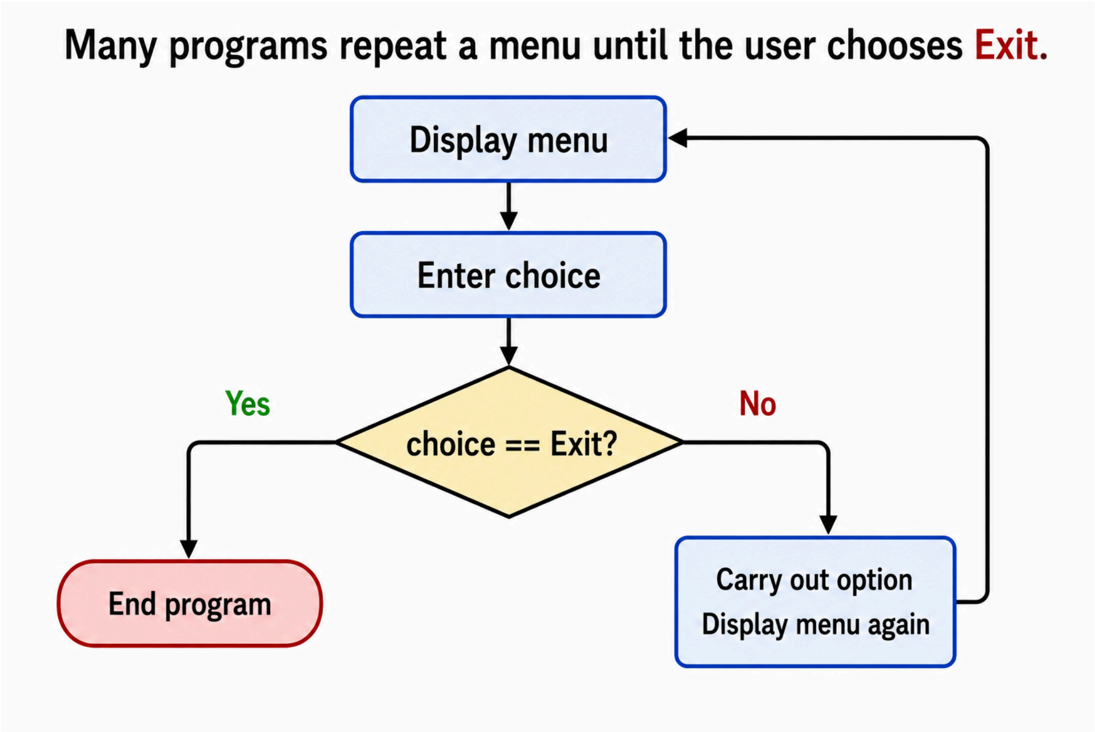
Use a loop when the user must enter valid data.



5.2 Sentinel loop



5.3 Menu loop


Many programs repeat a menu until the user chooses Exit.
Display menu
Enter choice
choice == Exit?
Yes
No
End program
Carry out option
Display menu again

Outline

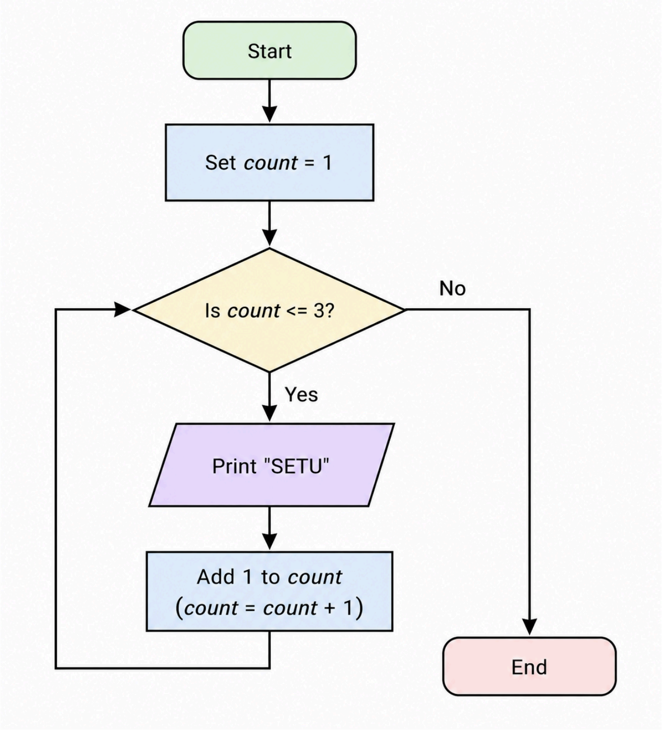
6.1 Activity: find the four parts

For the flowchart, identify:

1. Starting value
2. Condition
3. Repeated action
4. Update or stopping action

This prepares you to write the Python version in the next lecture.

6.2 Flowchart



6.3 Predict the output

Flowchart summary:

- Set `count = 1`
- While `count <= 3`
- Print `"SETU"`
- Add 1 to `count`

Question: what is printed?

6.4 Summary

- Iteration means repetition.
- Flowcharts help us plan loops before writing code.
- Selection asks once; iteration asks repeatedly.
- Every loop needs a condition and a way to stop.
- Missing updates can cause infinite loops.

Next: implementing these ideas in Python using `while` loops.

Thanks for Watching - Any questions?